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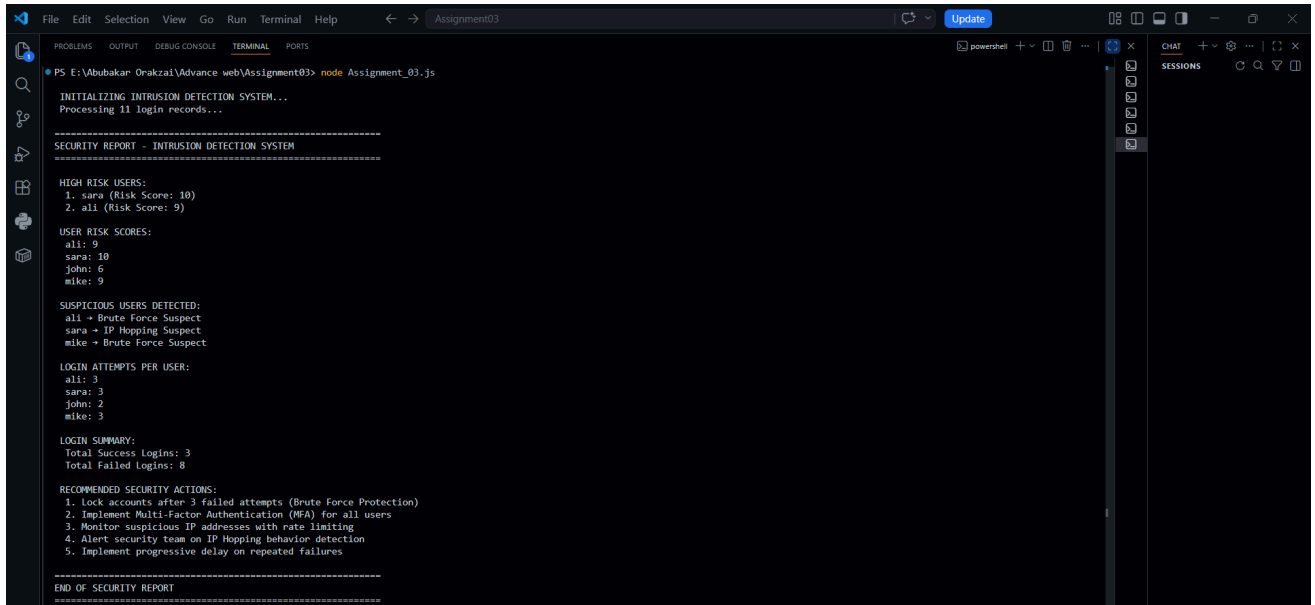
Subject : Advance Web

Assignment: 03

Submitted to : Sir Noman Saleem

Github link of the assignment:

Screenshot containing the output of the code



```
PS E:\Abubakar Orakzai\Advance web\Assignment03> node Assignment_03.js
INITIALIZING INTRUSION DETECTION SYSTEM...
Processing 11 login records...

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SECURITY REPORT - INTRUSION DETECTION SYSTEM
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HIGH RISK USERS:
1. sara (Risk Score: 10)
2. ali (Risk Score: 9)

USER RISK SCORES:
ali: 9
sara: 10
john: 6
mike: 9

SUSPICIOUS USERS DETECTED:
ali -> Brute Force Suspect
sara -> IP Hopping Suspect
mike -> Brute Force Suspect

LOGIN ATTEMPTS PER USER:
ali: 3
sara: 3
john: 2
mike: 3

LOGIN SUMMARY:
Total Success Logins: 3
Total Failed Logins: 8

RECOMMENDED SECURITY ACTIONS:
1. Lock accounts after 3 failed attempts (Brute Force Protection)
2. Implement Multi-Factor Authentication (MFA) for all users
3. Monitor suspicious IP addresses with rate limiting
4. Alert security team on IP Hopping behavior detection
5. Implement progressive delay on repeated failures

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END OF SECURITY REPORT
```

Detection Logic and Real-World Relevance in Web Security Systems

Detection Logic

The system works in three simple steps. First, it reads all login records and counts how many logins succeeded, how many failed, and how many times each user tried to log in. Second, it finds suspicious users using two rules. Rule 1: If a user fails to log in 3 or more times, they are a "Brute Force Suspect." Rule 2: If a user logs in

from different IP addresses, they are an "IP Hopping Suspect." Third, it gives each user a risk score. Failed logins add 2 points, successful logins add 1 point, and using different IPs adds 3 points. Finally, it shows the top 2 highest risk users and suggests security actions like locking accounts or adding multi-factor authentication.

Real-World Relevance in Web Security Systems

These detection methods are used by real companies every day. Brute force attacks are dangerous because hackers use computer programs to guess passwords thousands of times per second. If they guess correctly, they can steal money, personal data, or damage the system. IP hopping is suspicious because normal users log in from the same places, like home or work. If someone logs in from many different IP addresses quickly, it often means their password was stolen and many hackers are sharing it. Real websites like Gmail, Facebook, and online banks use these exact rules. They lock accounts after too many failed attempts and alert users about new or suspicious locations. This mini system shows how real security works to protect web applications from hackers.